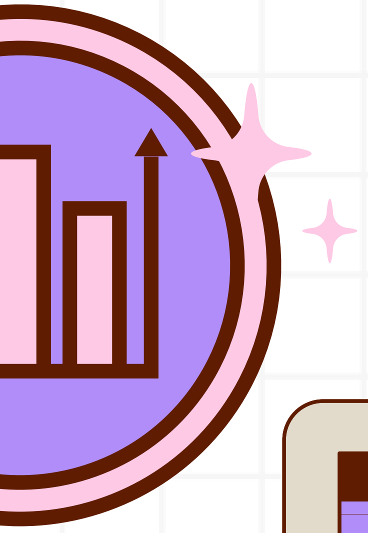
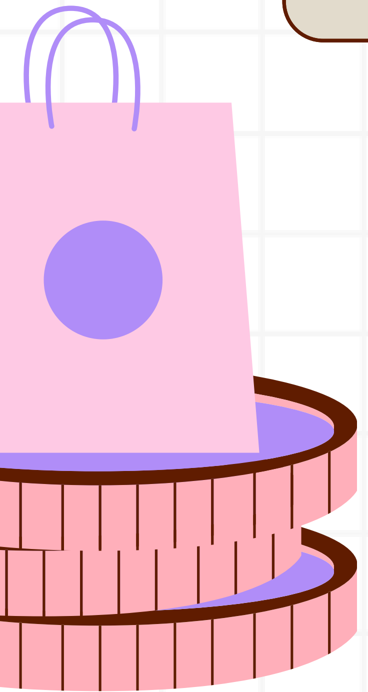


E-Commerce Performance & Customer Experience Analysis

Tracking Business Performance

Presented By: Saurabh Rai



E-Commerce Analytics

Analyzing Sales, Customers & Business Performance

Using SQL, this case study explores 5,000 e-commerce transactions to identify key trends across sales, customers, discounts, regions, delivery, and ratings. The analysis converts raw transactional data into meaningful business insights and recommendations.

Retrieve the total number of orders placed in the dataset

```sql
select count(*) as Total_Orders
from dbo.ECommerce_Sales_Analytics
```

Results | Messages

	Total_Orders
1	5000

“Analyzed the dataset to determine the total number of orders placed”.

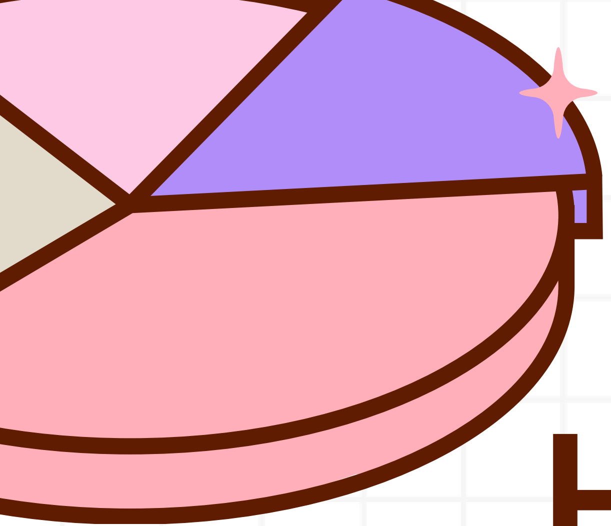
Calculate the total revenue generated from all sales.

```sql
select round(sum(revenue),2) as Total_Revenue
from dbo.ECommerce_Sales_Analytics;
```

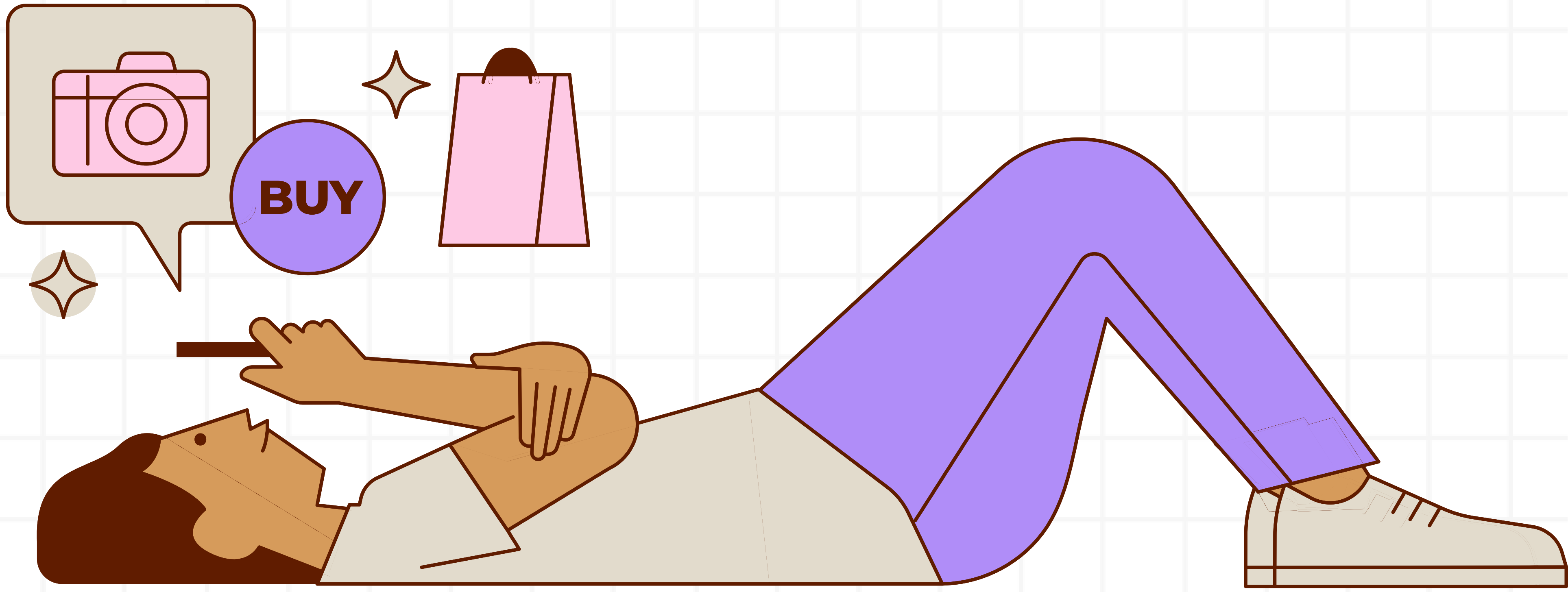
Results | Messages

	Total_Revenue
1	5109775.74

“Evaluated overall revenue performance from total sales.”



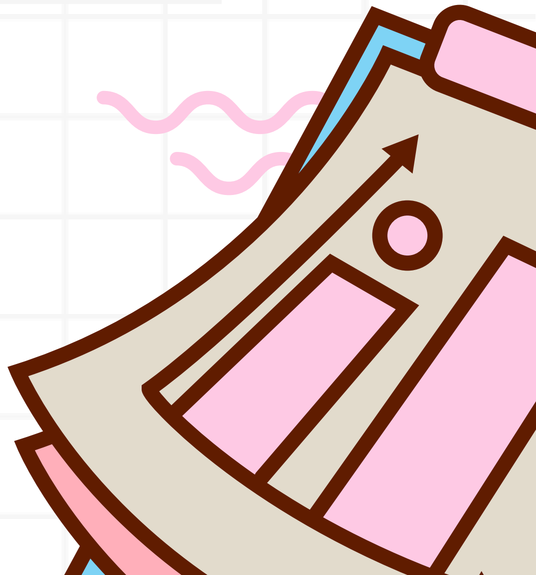
Highest-priced product item



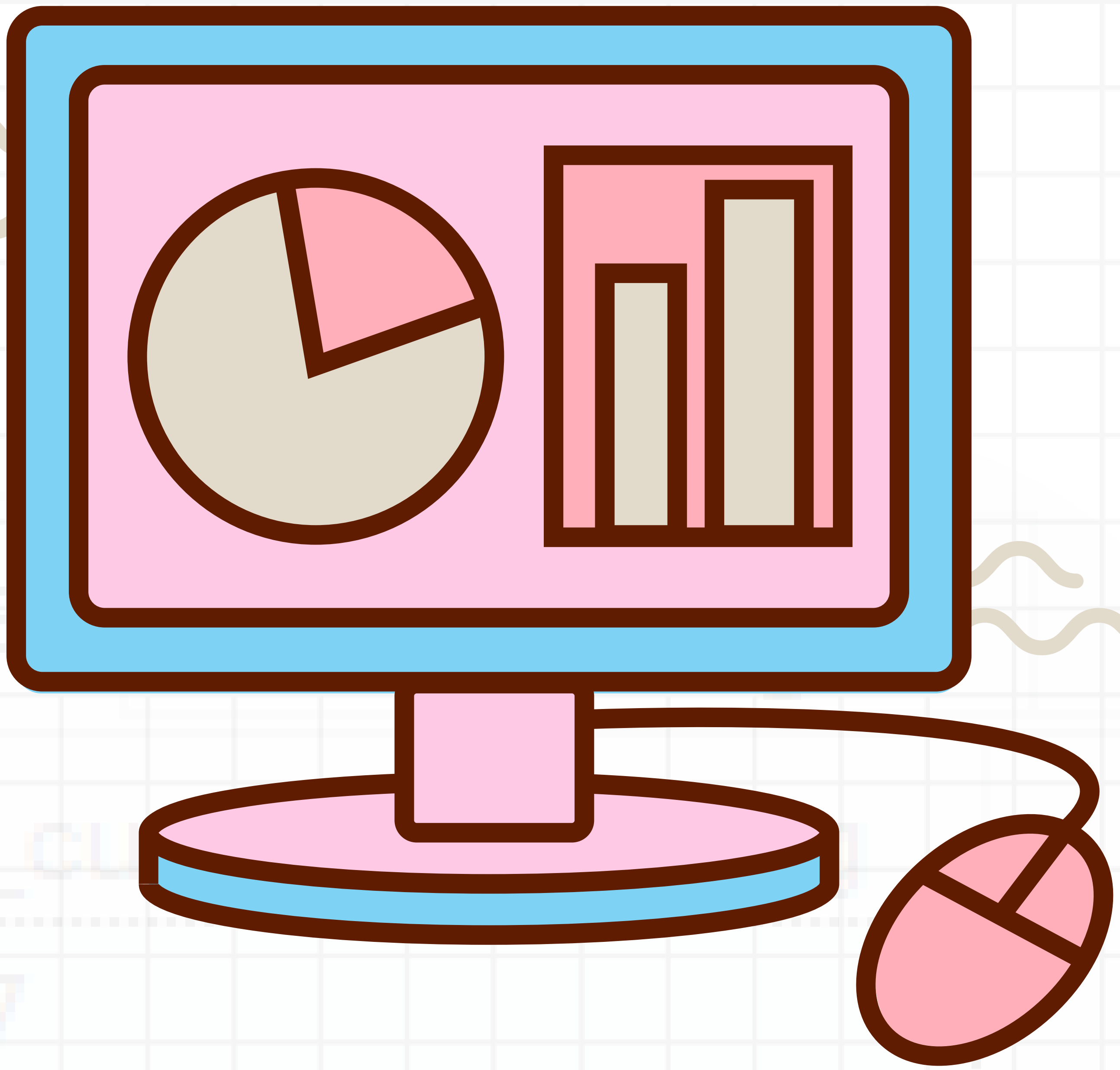

```sql
select TOP 1
order_id,
product_category,
round(unit_price,2) as max_unit_price
from dbo.ECommerce_Sales_Analytics
Order by unit_price desc;
```

Results | Messages

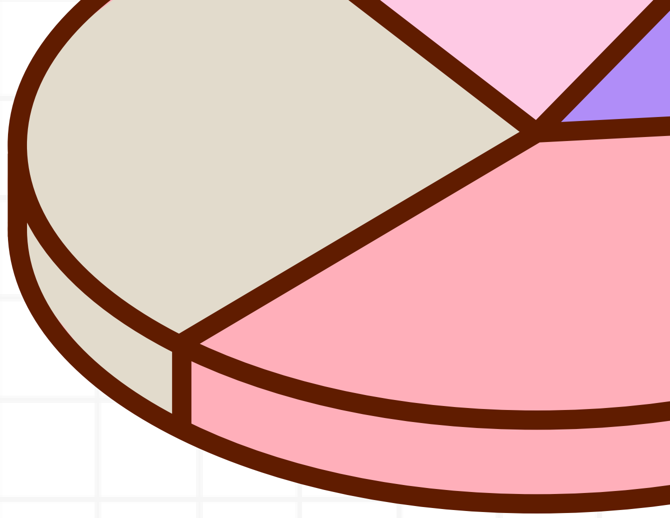
	order_id	product_category	max_unit_price
1	12702	Electronics	599.96



“Analyzed the product portfolio to identify the highest-priced item.”



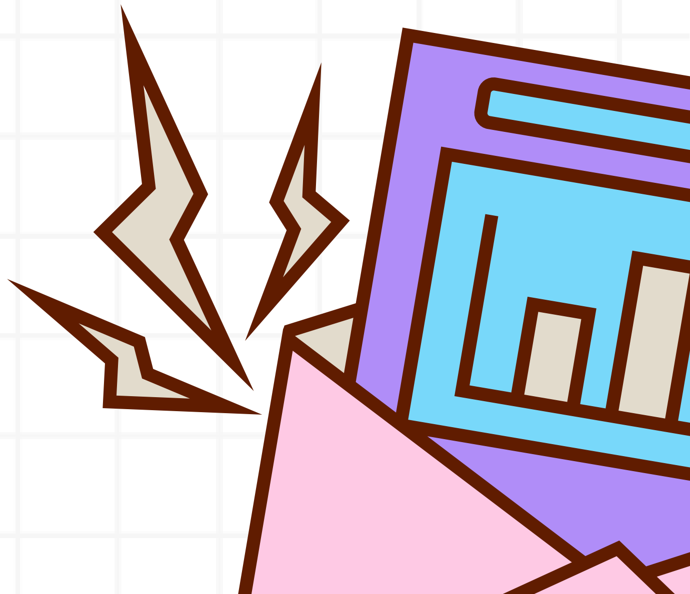
Average customer rating across all orders



Results | Messages

	avg_customer_rating
1	2.97

```sql
select
round(avg(customer_rating),2) as avg_customer_rating
from dbo.ECommerce_Sales_Analytics;
```



“Analyzed overall customer satisfaction based on average ratings.”

Revenue & AOV by Product Category

```sql
select product_category,
round(sum(revenue),2) as total_revenue,
round(avg(revenue),2) as avg_revenue
from dbo.ECommerce_Sales_Analytics
group by product_category
order by total_revenue desc;
```

Results | Messages

product_category	total_revenue	avg_revenue
Electronics	1829899.22	1029.77
Clothing	1531931.72	1000.61
Home	982083.92	1013.5
Beauty	765860.88	1059.28

“Analyzed revenue generation and average order value across product categories.”

Order Distribution by Payment Method

```sql
select
payment_method,
count(order_id) as total_order ,
round(sum(revenue),2) as total_revenue
from dbo.ECommerce_Sales_Analytics
group by payment_method
order by total_revenue desc;
```

Results Messages

	payment_method	total_order	total_revenue
1	Card	2270	2366248.2
2	COD	1774	1788408.44
3	Wallet	956	955119.1

“Analyzed order distribution across different payment methods.”

Monthly Revenue Trend

```sql
select
month(order_date) as order_month,
year(order_date) as order_year,
round(sum(revenue),2) as total_revenue
from dbo.ECommerce_Sales_Analytics
group by year(order_date), month(order_date)
order by year(order_date) asc, month(order_date) asc;
```

Results | Messages

	MoM	YoY	total_revenue
1	1	2022	35624.89
2	2	2022	29908.89
3	3	2022	30309.28
4	4	2022	30781.96
5	5	2022	41629.24
6	6	2022	29847.33
7	7	2022	28686.24
8	8	2022	29186.45
9	9	2022	28791.07
10	10	2022	29265.55
11	11	2022	32329.49
12	12	2022	25463.91
13	1	2023	37795.82
14	2	2023	30980.36
15	3	2023	26342.74
16	4	2023	26102.11
17	5	2023	32482.46
18	6	2023	35082.65
19	7	2023	27788.64
20	8	2023	30072.53
21	9	2023	32976.32
22	10	2023	29299.98
23	11	2023	30161.52
24	12	2023	26830.66

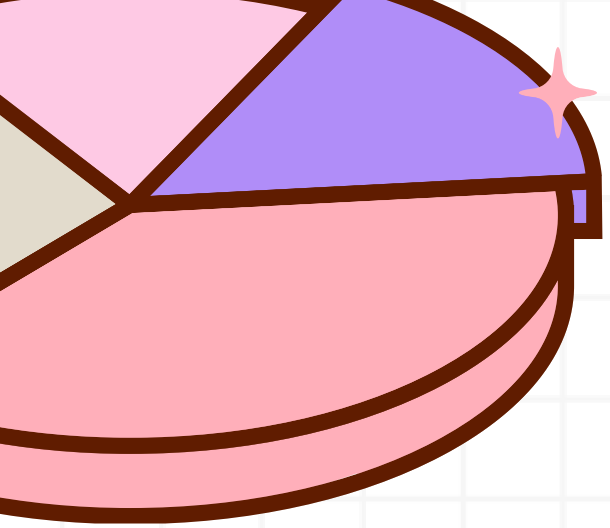
“Analyzing monthly revenue trends to evaluate sales performance over time.”

Payment Methods with Above-Average AOV

	payment_method	avg_revenue
1	Card	1042.4

```
select
payment_method,
round(avg(revenue),2) as avg_revenue
from dbo.ECommerce_Sales_Analytics
group by payment_method
having avg(revenue) > (select avg(revenue) from dbo.ECommerce_Sales_Analytics);
```

“Analyzed payment methods with above-average order values to identify higher-value purchasing patterns.”

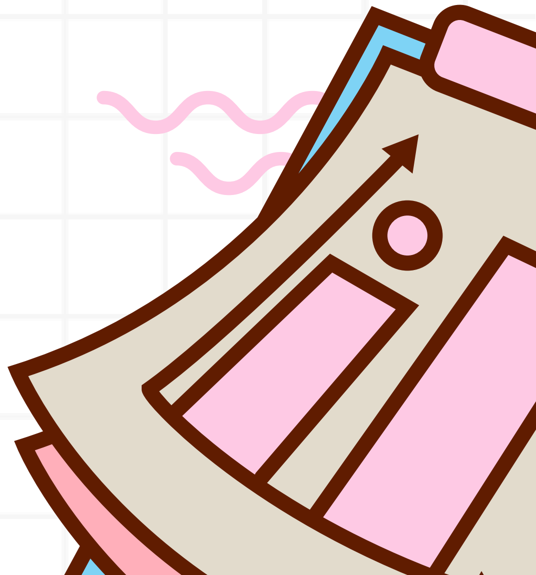


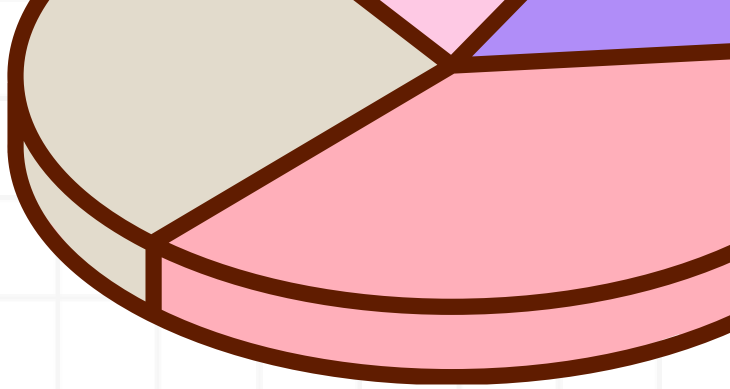
Revenue Contribution by Product Category

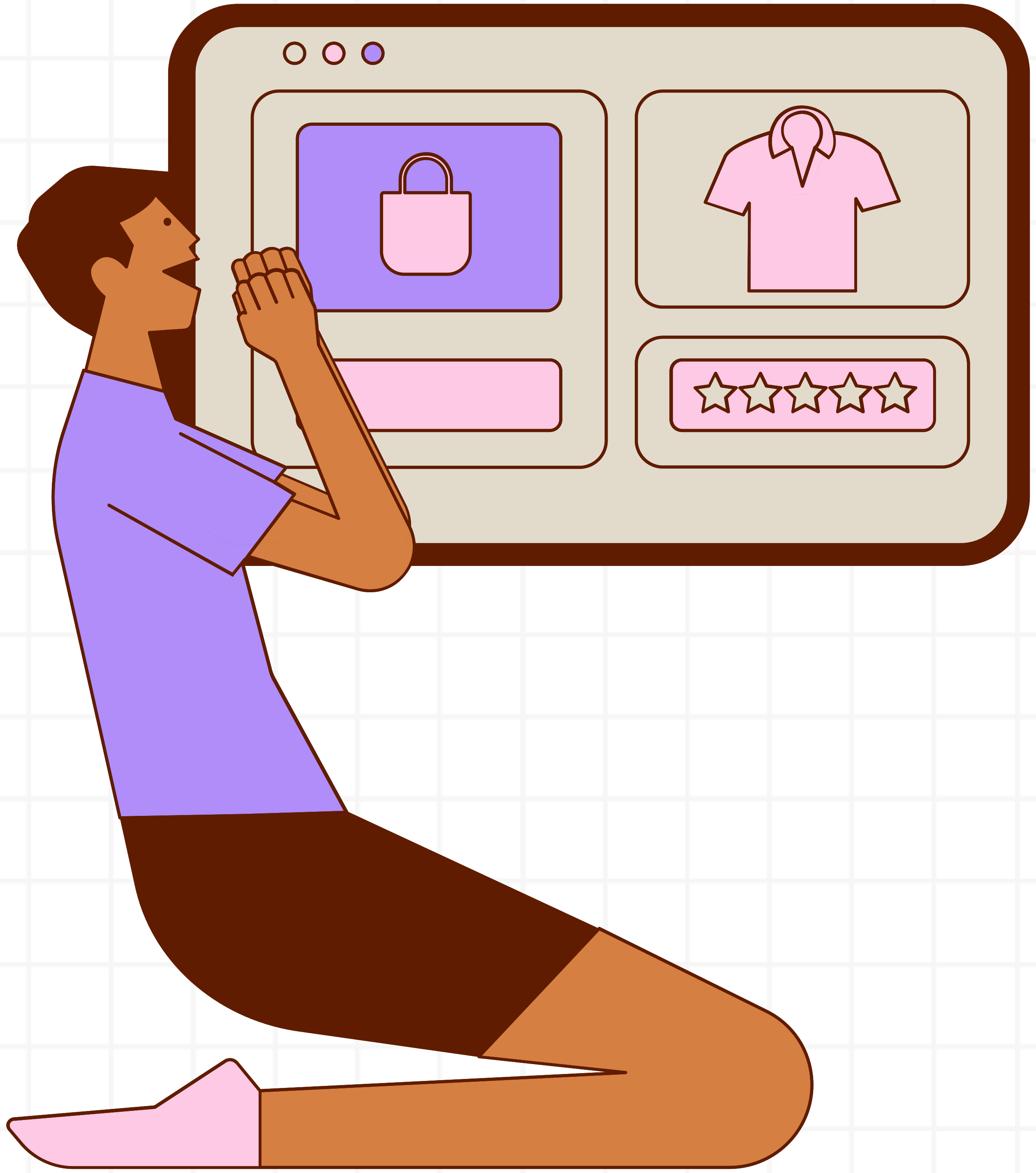
	product_category	grand_total_revenue	pct_contribution
1	Beauty	5109775.73903275	14.99
2	Home	5109775.73903275	19.22
3	Clothing	5109775.73903275	29.98
4	Electronics	5109775.73903275	35.81

```sql
select product_category ,
sum(per_category) over() as grand_total_revenue,
round(per_category / sum(per_category) over() * 100,2)  as pct_contribution
from(
select
product_category,
sum(revenue) as per_category
from dbo.ECommerce_Sales_Analytics
group by product_category) t
order by  pct_contribution;
```

“Analyzed each product category’s contribution to overall revenue.”







Thank You So Much

SQL Sales Analysis Project

Saurabh Rai

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